

Lab 3 – Component Modelling & Architectural Pattern Selection

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System: Alumni Mentorship & Mock Interview Platform

Architecture Selection

We chose **Layered Architecture** for the Alumni Mentorship & Mock Interview Platform. The system is divided into the Presentation Layer, Business Layer, and Data Layer. The Presentation Layer contains the Web User Interface, the Business Layer contains the Mentor Discovery, Session Manager, and Meeting Service components, and the Data Layer contains the Platform Database.

Reasons for Architectural Choice

1. Clear separation of responsibilities:

The platform contains distinct functions such as mentor discovery, session management, meeting-link generation, and data storage. Layered architecture separates the user interface, business logic, and data management, making the system easier to understand, develop, test, and maintain.

2. Suitable for the platform's centralized workflow:

Users interact with a single web interface while the business components coordinate mentor discovery and interview-session management. These components access a common database, making a layered structure appropriate without introducing the additional deployment and communication complexity of microservices.

Security Advantage

Layered architecture improves security by preventing the user interface from directly accessing the database. User requests first pass through the business layer, where authentication, authorization, and input validation can be performed before any data is retrieved or modified. This reduces unauthorized access to mentor, student, and session information.

Performance Benefit

The architecture allows frequently used operations, such as mentor searches and session information retrieval, to be handled efficiently by dedicated business components. Database access is centralized through the data layer, reducing unnecessary direct database interactions and making query optimization easier.